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Model theory

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## *Model Theory*

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## INTRODUCTION

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Should I begin by defining ‘model theory’? This might be unsafe – do the readers get their money back if the definition doesn’t match the contents? But here is an attempt at a definition: Model theory is the study of the construction and classification of structures within specified classes of structures.

A ‘specified class of structures’ is any class of structures that a mathematician might choose to name. For example it might be the class of abelian groups, or of Banach algebras, or sets with groups which act on them primitively. Thirty or forty years ago the founding fathers of model theory were particularly interested in classes specified by some set of axioms in first-order predicate logic – this would include the abelian groups but not the Banach algebras or the primitive groups. Today we have more catholic tastes, though many of our techniques work best on the first-order axiomatisable classes. One result of this broadening is that model theorists are usually much less interested than they used to be in the syntactical niceties of formal languages – if you want to know about formal languages today, you should go first to a computer scientist.

‘Construction’ means building structures, or sometimes families of structures, which have some feature that interests us. For example we might look for a graph which has very many automorphisms, or a group in which many systems of equations are solvable; or we might want a family of boolean algebras which can’t be embedded in each other. ‘Classifying’ a class of structures means grouping the structures into subclasses in a useful way, and then proving that every structure in the collection does belong in just one of the subclasses. An archetypal example from algebra is the classification of vector spaces over a fixed field: we classify them by showing that each vector space has a unique dimension which determines it up to isomorphism. Model



theory studies construction and classification in a broad setting, with methods that can be applied to many different classes of structure.

To give this book a shape, and to make it different from books which other people might write, I chose to concentrate on construction rather than classification. But that was twelve years ago, when (encouraged by Paul Cohn) I first sat down to write this book. Since then spells as editor, deputy head, dean etc. have destroyed any schedule that I ever had for writing the book, and of course the subject has moved on. The result was that far too much material accumulated. Some of it I diverted into three other books. Slightly over one megabyte has been shunted off to a file named Reject – I say this to appease readers who are annoyed to find no mention of their favourite topics or papers. The rejected material covers model theory of fields, atomic compact structures and infinitary languages, among several other things. These are valuable topics, but I simply ran out of space. (Other topics are missing out of brute ignorance – for example constructive model theory.)

Of the three other books, one has already appeared under the title *Building models by games*. It was originally a part of the present Chapter 8 (and it included the material that Paul Cohn had asked to see). A second book provisionally has the title *Structure and classification*, and will include much more stability theory. But several authors have already been kind enough to refer to the present book for some items in stability theory, and so I have taken the subject far enough to include those items. The third book will develop quasivarieties and Horn theories. Since this material has become more important for specification languages and logic programming than it ever was for model theory, that book will be aimed more at computer scientists.

After all these adjustments, the present book still has the emphases that I originally intended, though there is a lot more in it than I planned. Nearly every chapter is designed around some model-theoretic method of construction. In Chapter 1 it is diagrams, Chapter 3 includes Skolem hulls, Chapter 5 discusses interpretations as a method of construction, Chapter 6 tackles elementary amalgamation, Chapter 7 discusses omitting types and the Fraïssé construction, Chapter 8 is about existential closure, Chapter 9 deals with products, Chapter 10 builds saturated structures by unions of chains, Chapter 11 is about the Ehrenfeucht–Mostowski construction. That leaves Chapters 2, 4 and 12: Chapter 2 covers essential background material on languages, while Chapters 4 and 12 contain some recent developments of a geometric kind which I included because they are beautiful, important or both.

In the fourth century BC there was a bizarre philosophical debate about whether thought goes in straight lines or circles. Aristotle very sensibly supported the straight line theory, because (he maintained) proofs are linear.

Plato said circles, for astrological reasons which I wouldn't even wish to understand. But writing this book has convinced me that, just this once, Plato was right. There is no way that one can sensibly cover all the material in the book so that the later bits follow from the earlier ones. Time and again the more recent or sophisticated research throws up new information about the basic concepts. The later sections of several chapters, particularly Chapters 4 and 5, contain recent results which depend on things in later chapters. I trust this will cause no trouble; there are plenty of signposts in the text.

It would be hopeless to try to acknowledge all the people who have contributed to this book; they run into hundreds. But I warmly thank the people who read through sections – either of the final book or of parts now discarded – and gave me comments on them. They include Richard Archer, John Baldwin, Andreas Baudisch, Oleg Belegadek, Jeremy Clark, Paul Eklof, David Evans, Rami Grossberg, Deirdre Haskell, Lefty Kreouzis, Dugald Macpherson, Anand Pillay, Bruno Poizat, Philipp Rothmaler, Simon Thomas. I also thank David Tranah of Cambridge University Press for his encouragement and patience.

I owe a particular debt to Ian Hodkinson. There is no chance whatever that this book is free from errors; but thanks to his eagle eye and sound judgement, the number of mistakes is less than half what it would otherwise have been. His comments have led to improvements on practically every page. I say no more about his generous efforts for fear of getting him into trouble with his present employers, who must surely regard this as time misspent.

Finally a dedication. If this book is a success, I dedicate it to my students and colleagues, past and present, in the field of logic. Many of them appear in the pages which follow; but of those who don't, let me mention here two thoughtful and generous souls, Geoffrey Kneebone and Chris Fernau, both now retired, who ran the logic group of London University at Bedford College when I first came to London. If the book is not a success, I dedicate it to the burglars in Boulder, Colorado, who broke into our house and stole a television, two typewriters, my wife Helen's engagement ring and several pieces of cheese, somewhere about a third of the way through Chapter 8.

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## NOTE ON NOTATION

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Some exercises are marked with an asterisk \*. This means only that I regard them as not the main exercises; maybe they assume specialist background, or they are very difficult, or they are off centre.

I assume Zermelo–Fraenkel set theory, ZFC. In particular I always assume the axiom of choice (except where the axiom itself is under discussion). I never assume the continuum hypothesis, existence of uncountable inaccessible etc., without being honest about it.

The notation  $x \subseteq y$  means that  $x$  is a subset of  $y$ ;  $x \subset y$  means that  $x$  is a proper subset of  $y$ . I write  $\text{dom}(f)$ ,  $\text{im}(f)$  for the domain and image of a function  $f$ . ‘Greater than’ means greater than, never ‘greater than or equal to’.  $\mathcal{P}(x)$  is the power set of  $x$ .

Ordinals are von Neumann ordinals, i.e. the predecessors of an ordinal  $\alpha$  are exactly the elements of  $\alpha$ . I use symbols  $\alpha$ ,  $\beta$ ,  $\gamma$ ,  $\delta$ ,  $i$ ,  $j$  etc. for ordinals;  $\delta$  is usually a limit ordinal. A cardinal  $\kappa$  is the smallest ordinal of cardinality  $\kappa$ , and the infinite cardinals are listed as  $\omega_0$ ,  $\omega_1$  etc. I use symbols  $\kappa$ ,  $\lambda$ ,  $\mu$ ,  $\nu$  for cardinals; they are not assumed to be infinite unless the context clearly requires it (though I have probably slipped on this point once or twice). Natural numbers  $m$ ,  $n$  etc. are the same thing as finite cardinals.

‘Countable’ means of cardinality  $\omega$ . An infinite cardinal  $\lambda$  is a **regular** cardinal if it can’t be written as the sum of fewer than  $\lambda$  cardinals which are all smaller than  $\lambda$ ; otherwise it is **singular**. Every infinite successor cardinal  $\kappa^+$  is regular. The smallest singular cardinal is  $\omega_\omega = \sum_{n < \omega} \omega_n$ . The **cofinality**  $\text{cf}(\alpha)$  of an ordinal  $\alpha$  is the least ordinal  $\beta$  such that  $\alpha$  has a cofinal subset of order-type  $\beta$ ; it can be shown that this ordinal  $\beta$  is either finite or regular. If  $\alpha$  and  $\beta$  are ordinals,  $\alpha\beta$  is the ordinal product consisting of  $\beta$  copies of  $\alpha$  laid end to end. If  $\kappa$  and  $\lambda$  are cardinals,  $\kappa\lambda$  is the cardinal product. The context should always show which of these products is intended.

Some facts of cardinal arithmetic are assembled at the beginning of section 10.4.

Sequences are well-ordered (except for indiscernible sequences in Chapter 11, and it is explicit there what is happening). I use the notation  $\bar{x}$ ,  $\bar{a}$  etc. for sequences  $(x_0, x_1, \dots)$ ,  $(a_0, a_1, \dots)$  etc., but loosely: the  $n$ th term of a sequence  $\bar{x}$  may be  $x_n$  or  $x(n)$  or something else, depending on the context, and some sequences start at  $x_1$ . Sequences of finite length are called **tuples**. The terms of a sequence are sometimes called its **items**, to avoid the ambiguity in the term ‘term’. A sequence is said to be **non-repeating** if no item occurs twice or more in it. If  $\bar{a}$  is a sequence  $(a_0, a_1, \dots)$  and  $f$  is a map, then  $f\bar{a}$  is  $(fa_0, fa_1, \dots)$ . The length of a sequence  $\sigma$  is written  $\text{lh}(\sigma)$ . If  $\sigma$  is a sequence of length  $m$  and  $n \leq m$ , then  $\sigma|n$  is the initial segment consisting of the first  $n$  terms of  $\sigma$ . The set of sequences of length  $\gamma$  whose items all come from the set  $X$  is written  ${}^\gamma X$ . Thus  ${}^2 2$  is the set of ordered  $n$ -tuples of 0’s and 1’s;  ${}^{<\gamma} X$  is  $\bigcup_{\alpha < \gamma} {}^\alpha X$ . I write  $\eta$ ,  $\zeta$ ,  $\theta$  etc. for linear orderings;  $\eta^*$  is the ordering  $\eta$  run backwards.

I don’t distinguish systematically between tuples and strings. If  $\bar{a}$  and  $\bar{b}$  are strings,  $\bar{a} \hat{\ } \bar{b}$  is the concatenated string consisting of  $\bar{a}$  followed by  $\bar{b}$ ; but often for simplicity I write it  $\bar{a}\bar{b}$ . There is a clash between the usual notation of model theory and the usual notation of groups: in model theory  $xy$  is the string consisting of  $x$  followed by  $y$ , but in groups it is  $x$  times  $y$ . One has to live with this; but where there is any ambiguity I have used  $x \hat{\ } y$  for the concatenated string and  $x \cdot y$  for the group product.

Model-theoretic notation is defined as and when we need it. The most basic items appear in Chapter 1 and the first five sections of Chapter 2.

‘I’ means I, ‘we’ means we.